



Panspermia?

This isn't what you think it is. Panspermia is the theory that life exists throughout the universe and that asteroids are its distributors

The NEO program

This program is an initiative by NASA which lists and catalogues all Near-Earth-Objects currently discovered and under observation

How will we deal with a massive asteroid?

A 400m diameter asteroid is enough to wipe out the entire human race! What will we do if one such asteroid comes hurtling towards our planet?

Anirudh Regidi
anirudh.regidi@thinkdigit.com

At last count, there were over a thousand near-Earth asteroids (NEOs) that we know of, which is assumed to be only about 20 per cent (or lesser) of the actual number. Now consider the following statistics, the Earth is hit by a 45m asteroid about once every 100 years, a 300m asteroid every 10,000 years, a 1-6km asteroid every 1,000,000 years, and a 6km+ asteroid every 100,000,000 years.

Of these, the 45m asteroid can level a small town even if it blows up in the atmosphere, a 300m asteroid can level the UK or vaporize billions of tonnes of water if it lands in the oceans. A 300m asteroid by itself will not cause extinction, but the vaporized sea water can create such dense clouds that we probably won't see the sun again till the Earth freezes over and we're mostly extinct! This from an asteroid the size of an aircraft carrier! A 1km+ diameter asteroid will wipe out all life on Earth and cause massive tectonic shifts, a 6km+ asteroid, with sufficient speed, will even knock the Earth out of orbit and literally melt a vast section of the Earth's crust with the sheer pressure of the impact. Such an impact might even render the Earth unviable for life for millions of years (the dinosaurs might have died to an asteroid like this).

Tracking these monsters

If this wasn't scary enough, did you know that in 1989, a massive asteroid missed us by just 6 hrs and we didn't know about it



The impact of a super-massive asteroid. This one is the size of a small moon!

till long after it crossed the Earth's orbit, or that such events happen once every 2-3 years and we are completely oblivious? You might think that with the vast number of electronic eyes keeping an eye on space, something as large as an asteroid will not go unnoticed; but the problem with space is that it is vast, mind-bogglingly vast. Everything happens at an incredible scale, but also at an incredibly slow rate (in human terms). An asteroid might be hurtling towards us at over 25,000 km/hr, but it would still take years to reach us even from Jupiter. If you consider that distance (more than 200 million kms), pin-

pointing a tiny rock (at that scale) in space is an almost impossible task, and even if we had a thousand such asteroids hurtling towards us, we would be none the wiser unless we knew exactly where to look.

The task is much easier now of course, advanced radar and fancy satellites are helping us, not to mention the vast supercomputing power that is being utilized just to track all the known objects in the sky. Nowadays we could, in theory, plot the course and predict the time of impact of an Earth-bound asteroid by as much as a few decades in advance. In a worst case scenario, that time would be less than a decade, but we'll still have a few years to respond.

What is an asteroid?

We've always thought of asteroids as massive chunks of rock, the primordial "ooze" from which the planets were formed and that is partly true. The truth is that a lot of the asteroids that we are aware of are actually just amalgamations of debris, vast clumps of "boulders" and dust and are mostly just empty volume. If such an asteroid were to hit the Earth,



An object of destruction or source of life?